

1 Introduction

Quantum many-body systems are difficult to study because the state space grows exponentially with the number of particles. Even for a spin- $\frac{1}{2}$ chain of length L , a pure state is a complex-valued function

$$\psi : \{-1, +1\}^L \rightarrow \mathbb{C},$$

so that computing the ground state of a given Hamiltonian becomes a high-dimensional variational problem. Classical methods such as exact diagonalization and DMRG are very effective in some cases, but they can be hard to generalize, especially in higher dimensions or for more complex interactions. This motivates the search for wavefunction approach that can exploit the structure of lattice models.

In this project, we study spin models such as the Ising and Heisenberg chains as a testbed for *neural quantum states*. Our goal is to approximate ground-state wavefunctions with a neural network and to train this network to minimize the energy. Spin configurations on a one-dimensional chain can be written as

$$\sigma = (\sigma_1, \dots, \sigma_L) \in \{-1, +1\}^L,$$

and in two dimensions as an $L_x \times L_y$ grid in $\{-1, +1\}^{L_x \times L_y}$. These configurations live on a regular lattice with local interactions and translation symmetry, which makes them well matched to convolutional neural networks (CNNs). CNNs apply small filters that are shared across sites, so that the same local pattern is recognized anywhere in the lattice. This weight sharing yields efficiency and often approximate translational equivariance, while depth allows the network to build up from short-range to longer-range correlations.

Our approach is to represent spin configurations as one- or two-dimensional arrays and feed them into a CNN with one- or two-dimensional convolutional layers. The CNN defines a parametrized function

$$\psi_\theta : \{-1, +1\}^L \rightarrow \mathbb{C},$$

which we interpret as an ansatz for the ground-state wavefunction. We train ψ_θ using variational Monte Carlo: configurations are sampled, local energies are estimated, and the parameters θ are updated to reduce the estimated ground-state energy. We first benchmark this CNN-based neural quantum state on one-dimensional Ising and Heisenberg models against ground-state energies, and then discuss how the same architecture can be extended to more challenging lattice systems.

2 Background: Convolutional Neural Networks

Convolutional neural networks (CNNs) are neural networks designed for data on a grid (e.g., images or spin configurations). Like standard feedforward networks, they consist of layers with learnable weights and biases trained by gradient descent. Their key architectural idea is to exploit two common properties of such data: *locality* (nearby sites interact more strongly than distant ones)

and *translation symmetry* (the same local pattern can appear anywhere in the grid).

2.1 Convolution and feature maps

Unlike fully connected layers, a convolutional layer uses *local filters* (kernels). For a 2D input $x \in \mathbb{R}^{H \times W}$ (e.g., a spin configuration on an $H \times W$ lattice), a single filter is a small matrix

$$K \in \mathbb{R}^{k_h \times k_w}, \quad k_h \ll H, \quad k_w \ll W.$$

The convolution of x with K produces an output z defined by

$$z_{i,j} = \sum_{u=0}^{k_h-1} \sum_{v=0}^{k_w-1} K_{u,v} x_{i+u, j+v},$$

for all positions (i, j) where the filter fits inside the input. Intuitively, we slide K across the grid and at each location compute a weighted sum of the local patch. Using several filters $K^{(1)}, \dots, K^{(C_{\text{out}})}$ in parallel yields C_{out} *feature maps*, each responding to a different local pattern.

After this linear operation, a pointwise nonlinear σ ,

$$\sigma(t) = \max\{0, t\},$$

is applied to obtain activations $h_{i,j} = \sigma(z_{i,j})$. Stacking multiple convolution–activation layers builds a hierarchy of features: early layers detect simple local patterns, whereas deeper layers combine them into more complex structures.

Because the same filter K is reused at all spatial locations, the number of parameters in a convolutional layer depends only on the filter size (e.g., $k_h k_w$ per filter) and not on the overall input size $H \times W$. This *weight sharing* makes CNNs much more parameter-efficient than fully connected networks on high-dimensional grids and yields approximate translation equivariance: if the input is shifted, the feature maps shift in the same way (up to boundary effects).

2.2 Pooling, depth, and overall architecture

CNNs often include *pooling* layers in addition to convolutions. For example, max pooling with window size 2×2 replaces each non-overlapping 2×2 block of activations by its maximum. Pooling reduces spatial resolution, lowers computational cost, makes the representation less sensitive to small translations and noise, and effectively enlarges the receptive field of deeper neurons.

Intuitively, standard CNN for supervised learning looks like:

$$(\text{convolution} + \text{nonlinearity}) \rightarrow (\text{optional pooling}),$$

followed by one or more fully connected layers that map the final feature maps to the desired output (e.g., class scores). Given training data $\{(x^{(n)}, y^{(n)})\}_{n=1}^N$, the network defines a function f_θ with

parameters θ (all filter weights and biases). These parameters are learned by minimizing

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{n=1}^N \ell(f_{\theta}(x^{(n)}), y^{(n)}),$$

using optimization methods through all layers.

3 Dataset and Exploratory Analysis

3.1 Dataset construction

The dataset contains 73,710 simulated quantum spin-chain configurations generated by exact diagonalization of one-dimensional Heisenberg models. For each instance we embed a chain with $N \in \{1, \dots, 12\}$ active spins into a $1 \times 1 \times 14$ dimension array, namely a 14 dimension input. The physical spins occupy sites 0 to $N - 1$ and take values in $\{-1, +1\}$; sites N to 13 are zero-padded to 0. Each configuration is paired with couplings (J_1, J_2) , where $J_1 \in \{1.0, 2.0, 3.0\}$ and $J_2 \in \{0.25, 0.50, 0.75, 1.0, 2.0\}$.

The dynamics are governed by the Ising-like Hamiltonian

$$H = -J_1 \sum_{i=1}^{N-1} \sigma_i \sigma_{i+1} - J_2 \sum_{i=1}^{N-2} \sigma_i \sigma_{i+2}, \quad (3.1)$$

with $\sigma_i \in \{-1, +1\}$. For each (J_1, J_2, N) we enumerate all 2^N spin configurations and diagonalize H , obtaining the ground-state energy E_0 and the associated eigenvector. The learning target is the ground-state eigenvector component associated with each spin configuration in the computational basis. Across the 9 (J_1, J_2) pairs this yields 8,190 samples per parameter pair, for a total of 73,710 examples.

3.2 Preprocessing

We keep spin configurations in their original discrete form $\{-1, 0, +1\}$, where 0 marks padded sites. Target coefficients are left unscaled, since their range is already bounded and physically meaningful.

Each configuration is reshaped into a 1×14 grid for input to a convolutional network. We include N , J_1 , and J_2 as additional scalar features so the model can use both local spin structure and global couplings.

We split the data into training, validation, and test sets using a stratified 70 -15 -15 split over (N, J_1, J_2) , producing 51,597 training, 11,056 validation, and 11,057 test samples. All summary statistics and plots below use the *training set only*; the validation set is reserved for model selection and the test set for final evaluation.

3.3 Exploratory data analysis

Figure 1 shows the training-set distribution of system size on a logarithmic scale. Counts grow roughly as 2^N , so chains with $N \geq 10$ dominate the data. Small systems are therefore underrepresented, which is worth keeping in mind when interpreting per- N performance.

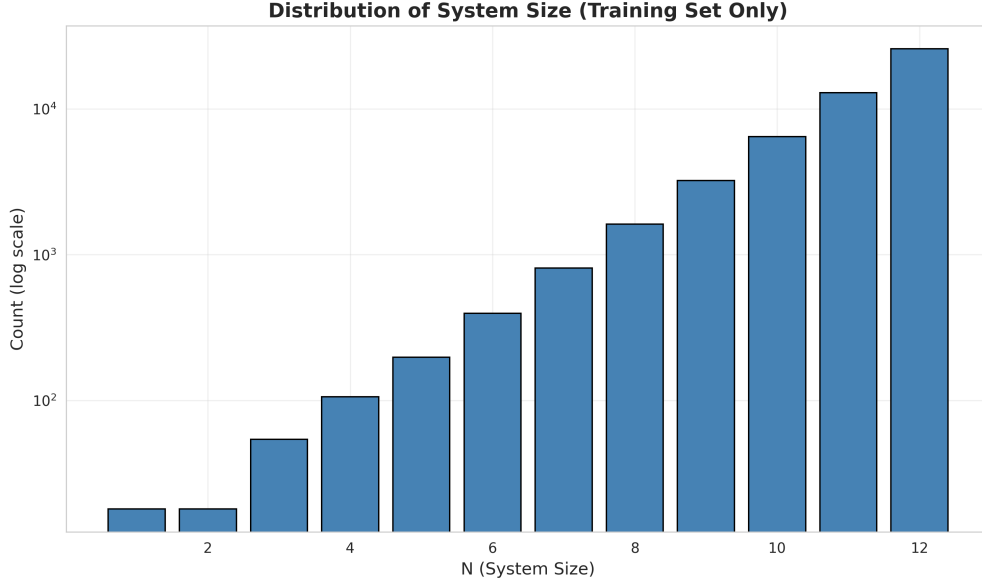


Figure 1: Distribution of system sizes N (log scale)

The ground-state eigenvectors are extremely sparse. In Figure 2, the left panel shows all training coefficients with a pronounced spike at zero; about three quarters of entries are exactly zero. The right panel restricts to non-zero coefficients, which cluster tightly around zero with relatively light tails.

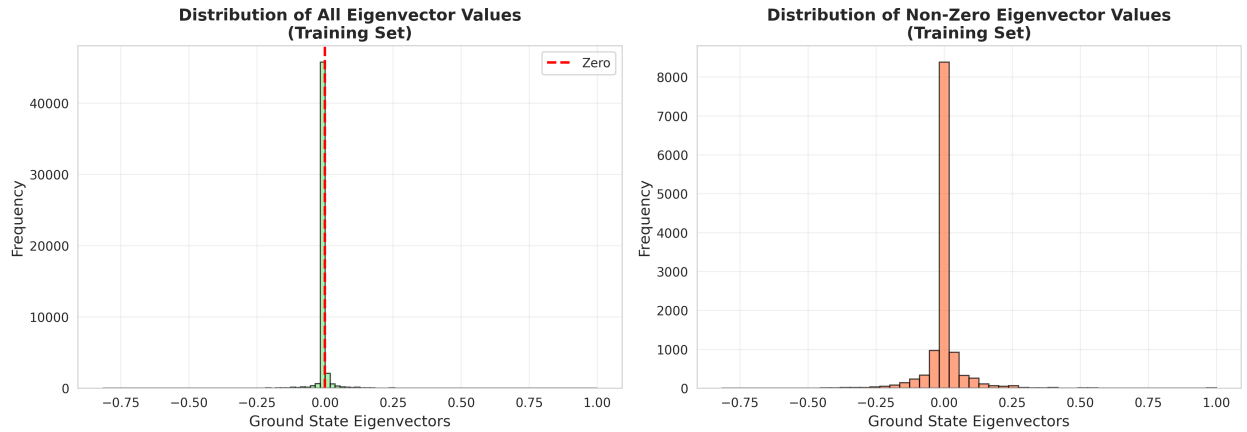


Figure 2: Distribution of ground-state eigenvector coefficients. Left: all coefficients, with a large spike at zero. Right: non-zero coefficients only, showing a narrow band of amplitudes around zero.

We also examine the ground-state energy E_0 . The left panel of Figure 3 plots E_0 against N for

all (J_1, J_2) , along with the mean energy at each size. The mean decreases almost linearly with N ; a least-squares fit yields

$$\mathbb{E}[E_0 | N] \approx -0.70 N + 0.07,$$

with correlation $r \approx -0.983$ on the training set. The right panel shows the marginal distribution of energies, which is broad and multi-modal across coupling regimes but centered around -7.6 .

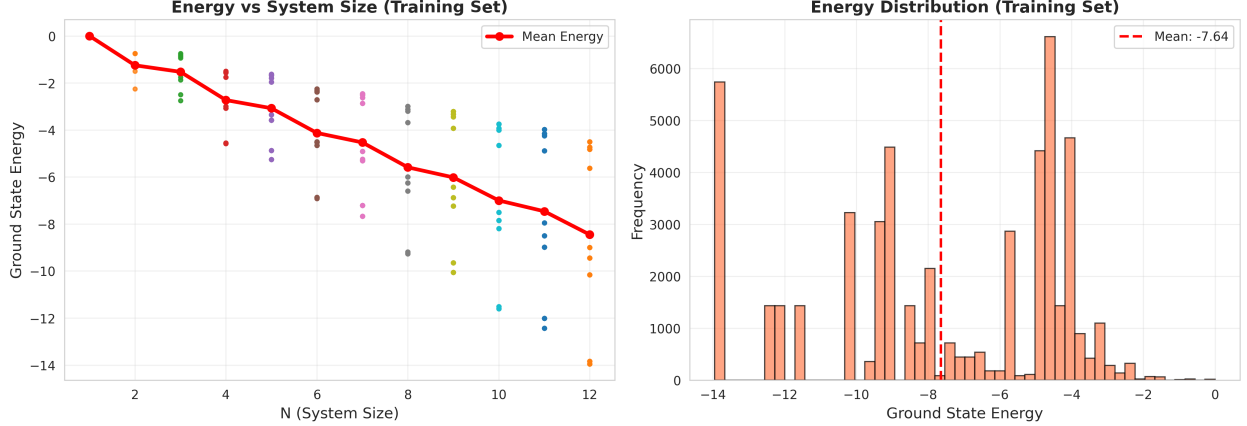


Figure 3: Energy statistics in the **training set**. Left: ground-state energy E_0 versus system size N with mean energy (red curve). Right: histogram of E_0 across all training examples.

Finally, simple correlations highlight why a non-linear model is needed. Pearson correlations between the input features (N, J_1, J_2) and the target coefficient are all smaller than 0.04 in magnitude, while E_0 is strongly correlated with both N and the couplings. Energy is almost linear in the inputs; individual eigenvector entries are not. This gap motivates using a convolutional neural network to capture the non-linear structure of the ground state.

4 Physical Model and Ground-State Physics

Our experiments are based on the one-dimensional J_1 - J_2 Ising chain already introduced in (3.1). Here we give a brief physical interpretation of this model and of the observables that our neural networks are implicitly learning.

4.1 Coupling regimes and frustration

The Hamiltonian

$$H = -J_1 \sum_{i=1}^{N-1} \sigma_i \sigma_{i+1} - J_2 \sum_{i=1}^{N-2} \sigma_i \sigma_{i+2}, \quad (4.1)$$

with $\sigma_i \in \{-1, +1\}$, couples nearest and next-nearest neighbors with strengths J_1 and J_2 . For $J_1 > 0$ and $J_2 = 0$, the energy is minimized when all spins align, so the two fully ferromagnetic configurations are exact ground states. In this unfrustrated regime the energy landscape is simple:

one or two configurations dominate, and local updates that align neighbors always lower (or leave unchanged) the energy.

As soon as the two couplings favor incompatible patterns, the picture changes. For example, if $J_1 > 0$ prefers alignment of neighbors but the effective contribution of J_2 favors anti-alignment of spins at distance two, no configuration can simultaneously minimize every bond term. The system is then said to be *frustrated*: at least one interaction in each local neighborhood is “unhappy.” In such regimes the ground state can no longer be guessed from a single simple pattern; instead, the minimizing configuration depends sensitively on the ratio J_2/J_1 and on the chain length N . Small changes in the couplings can reorder the spectrum, so that different spin patterns become the unique or degenerate ground state.

This competition between J_1 and J_2 is what makes the mapping

$$(N, J_1, J_2, \sigma) \mapsto E_0(N, J_1, J_2)$$

non-trivial. Even though the Hamiltonian is diagonal in the computational basis, the identity of the minimal-energy configuration changes across parameter space in a way that is not captured by simple linear correlations, as seen in the exploratory analysis.

4.2 Few-body intuition

For small chains the structure of the spectrum can be understood explicitly. For $N = 2$ with only nearest-neighbor coupling, the four possible spin pairs decompose into two energy levels: the aligned states $(+, +)$ and $(-, -)$ have energy $-J_1$, while the anti-aligned states $(+, -)$ and $(-, +)$ have energy $+J_1$. The ground-state energy is therefore $E_0 = -J_1$, with a two-fold degeneracy.

For $N \geq 3$ and non-zero J_2 , the next-nearest-neighbor terms split these simple degeneracies. Configurations that were previously equivalent in energy can be pushed up or down depending on how many bonds of each type they satisfy. As N increases, the number of competing local patterns grows exponentially, and tracking which configuration minimizes the total

$$E(\sigma) = -J_1 \sum_i \sigma_i \sigma_{i+1} - J_2 \sum_i \sigma_i \sigma_{i+2}$$

by hand becomes infeasible. The exact diagonalization pipeline used to generate our dataset effectively automates this process, but the underlying physics is still determined by the local bond structure.

4.3 Observables, correlations, and neural representation

The energy can be decomposed into local bond contributions

$$\varepsilon_i^{(1)} = -J_1 \sigma_i \sigma_{i+1}, \quad \varepsilon_i^{(2)} = -J_2 \sigma_i \sigma_{i+2},$$

which are directly proportional to two-point correlators $\langle \sigma_i \sigma_{i+r} \rangle$ at distances $r = 1$ and 2 . Minimizing E therefore amounts to arranging spins so that these short-range correlations match the pattern favored by the couplings. In regions of parameter space where the preferred patterns conflict, the ground state must compromise, leading to domains, modulated order, or rapidly changing low-energy configurations as (J_1, J_2) vary.

From this perspective, the role of the neural network is clear. In the supervised setting, the CNN is trained to approximate the mapping from a spin configuration (together with J_1 and J_2) to the corresponding ground-state energy, which is a nonlinear functional of these local bond products. In the variational Monte Carlo setting, the same convolutional architecture is used as a wavefunction ansatz $\psi_\theta(\sigma)$ whose squared modulus defines a probability distribution over configurations. Sampling from $|\psi_\theta(\sigma)|^2$ emphasizes regions of configuration space with favorable local correlations; gradient-based updates to θ then adjust this distribution to further lower the expected energy. The fact that the Hamiltonian is built from local, translationally repeated terms is mirrored by weight sharing and small convolutional kernels, which makes CNNs a natural physics-informed choice for this class of spin models.

5 Supervised CNN Model

Before deploying the Unsupervised Variational Monte Carlo (VMC) framework, we established a supervised baseline to validate our Neural Network architecture. The goal was to determine if a Convolutional Neural Network (CNN) could implicitly learn the mapping from a spin configuration and Hamiltonian parameters to the ground-state energy, $f(\sigma, J_1, J_2) \rightarrow E_0$, without requiring the explicit computation of the full wavefunction.

5.1 Supervised CNN Architecture

We designed a custom 1D-CNN tailored to the lattice dimensionality ($L = 14$). A standard CNN input containing only spin data would fail to capture the varying interaction strengths inherent in the problem. To address this, we implemented a physics-informed input representation by constructing a tensor (3×14) that embeds the Hamiltonian rules directly alongside the quantum state. This tensor stacks the spin configuration with two broadcasted channels containing the nearest-neighbor (J_1) and next-nearest-neighbor (J_2) coupling constants, allowing the model to learn local energy contributions dependent on specific coupling strengths.

The network architecture consists of two 1D convolutional layers followed by a regression head. The first convolutional layer (kernel size $k = 3$, 16 filters) captures the immediate local spin environment. The second layer increases feature depth to 32 channels; crucially, stacking this second layer expands the network’s receptive field, allowing it to resolve complex correlations including the next-nearest neighbor (J_2) interactions. We then apply Global Average Pooling to the spatial dimension, aggregating local energy densities into a single feature vector. This step enforces translational invariance, ensuring the model predicts energy based on the consistency of physical rules

across the chain rather than absolute positions. Finally, the pooled vector is fed into a Multi-Layer Perceptron to regress the scalar ground-state energy.

5.2 Training and Metrics

We generated a dataset of 73,710 configurations using Exact Diagonalization (ED) to provide the ground truth energy E_{GT} . The data was split into Training (70%), Validation (15%), and Test (15%) sets to ensure rigorous evaluation.

The training objective was to minimize the **Mean Squared Error (MSE)** between the predicted energy and the exact ground truth:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (\hat{E}_{\theta}(x_i) - E_{GT}^{(i)})^2$$

The model was trained for 20 epochs with a batch size of 64 using the Adam optimizer (learning rate 0.001) to minimize the Mean Squared Error against ground truth energies. By targeting the scalar energy rather than the sparse high-dimensional eigenvectors, we avoided the numerical instability and "zero-collapse" issues common in wavefunction regression.

5.3 Results and Interpretation

The supervised baseline demonstrated rapid convergence (Fig. 4). The validation loss tracked closely with training loss, stabilising after 5 epochs.

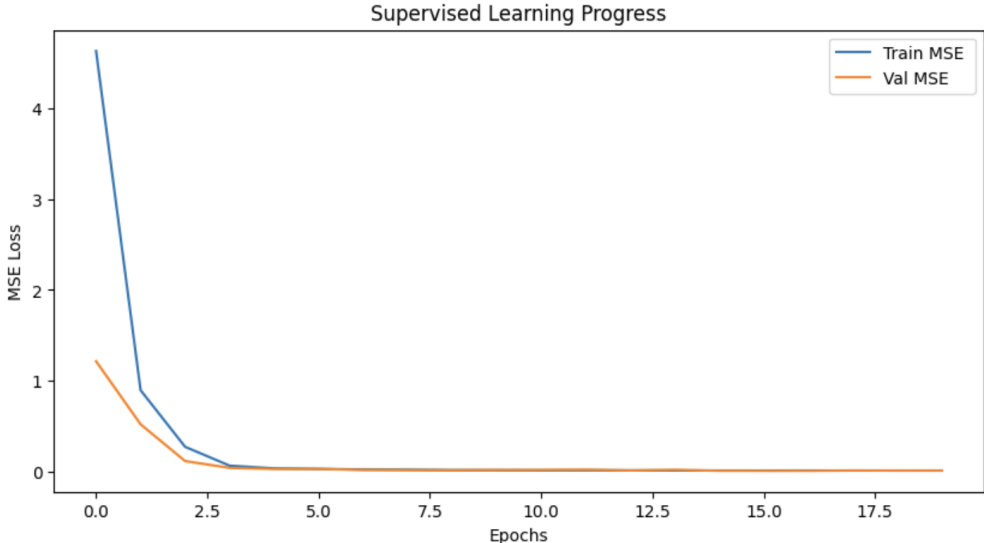


Figure 4: MSE Loss vs Epochs

Upon evaluation, the model achieved a final Test RMSE of 0.1399. Compared to the dataset's baseline standard deviation of 3.42, this represents a 24.4-fold improvement in accuracy, effectively capturing 99.8% of the variance in energy levels. With a Mean Absolute Error of 0.1180 against

a dataset mean energy of -7.64, the model demonstrates a relative error of approximately 1.5%, confirming that the CNN architecture can approximate exact diagonalization results with high precision.

This result validates the architecture as a suitable ansatz ψ_θ for the subsequent Unsupervised VMC phase, where the "ground truth" labels will be removed, and the network will be trained by minimizing the energy expectation directly.

6 Unsupervised CNN

6.1 Variational Sampling Principle

The core physics engine of our unsupervised model relies on the **Variational Principle**. In quantum mechanics, the expectation value of the Hamiltonian (the average energy) for any trial wavefunction $|\Psi_\theta\rangle$ acts as an upper bound on the true ground state energy E_0 :

$$\langle E \rangle = \frac{\langle \Psi_\theta | H | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} \geq E_0$$

Since the Hilbert space is too large to compute this integral exactly (scaling as 2^N states), we approximate it using statistical sampling. We first define a **Local Energy** $E_{loc}(S)$ for a specific spin configuration S , which involves comparing the wavefunction value of our current state with it's neighbors—states with a single $J1/J2$ interaction flipped in sign:

$$E_{loc}(S) = \sum_{S'} H_{S,S'} \frac{\psi_\theta(S')}{\psi_\theta(S)}$$

By generating a batch of samples S distributed according to the dynamic probability distribution $P(S) = |\psi_\theta(S)|^2$, our mcmc sampler `mcmc_step` guides the training samples toward a local minimum, allowing total system energy to be approximated as the statistical mean of the local energies:

$$\langle E \rangle \approx \frac{1}{M} \sum_{i=1}^M E_{loc}(S_i)$$

Minimizing this statistical quantity allows the neural network to "find" the ground state, and act as our error function. In our code, this estimation is found under `compute_local_energy_log` and `compute_energy_complex_robust`.

Consistent with the architectural principles outlined in Section 2.3 of Lange et al. (2024)(2), We utilized 1D Convolutional Neural Networks (CNNs) as the ansatz for the wavefunction $\Psi(S)$ due to the translational invariance of the spin chain.

6.2 Real-Valued CNN

This architecture outputs a single real scalar representing the log-amplitude of the wavefunction.

- **Structure:** Input $(1, N) \rightarrow \text{Conv1D} \rightarrow \text{ELU} \rightarrow \text{Conv1D} \rightarrow \text{Global Pooling} \rightarrow \text{Dense} \rightarrow \text{Output}$. ELU is chosen over ReLU to preserving gradient flow crucial for capturing subtle quantum correlations. Global Average Pooling is preferred over Maxpool because the total energy is an extensive property, whereas Max Pooling discards information by only keeping the strongest activation.
- **Wavefunction:** The scalar output represents the log-probability:

$$\Psi(S) = e^{\text{output}}$$

- **Limitation:** Since e^x is strictly positive, this model cannot represent wavefunctions with varying signs naturally. We compensated by hard-coding the **Marshall Sign Rule** (multiplying the output by -1 for odd parity states), however this model fails for strongly frustrated systems, and fails to internalize important physical dynamics.

6.3 Complex-Valued CNN

This architecture outputs two real scalars for every state, inspired by the review from Lange et al. (2024)(1).

- **Structure:** Similar feature extraction to the Real CNN, but the final Dense layer splits into two distinct heads:
 1. **Log-Amplitude Head:** Outputs $\ln|A|$.
 2. **Phase Head:** Outputs phase angle ϕ .
- **Wavefunction:** The network represents the state as:

$$\Psi(S) = e^{\ln|A| + i\phi}$$

- **Advantage:** This model can learn to rotate the phase ϕ of specific states to be π (equivalent to a -1 sign) or any other angle. It allows the network to automatically discover the sign structure required to minimize energy in frustrated systems.

6.4 Results and Interpretations

We performed a series of experiments to validate the efficacy of Neural Quantum States across different system sizes and architectures.

Hyperparameter Tuning To optimize the Complex-Valued CNN for larger systems ($N \geq 10$), we performed a grid search over kernel sizes, and layer depths. The most significant finding was the impact of the convolution kernel size.

We found that a **kernel size of 5** consistently outperformed smaller kernels ($k = 3$).

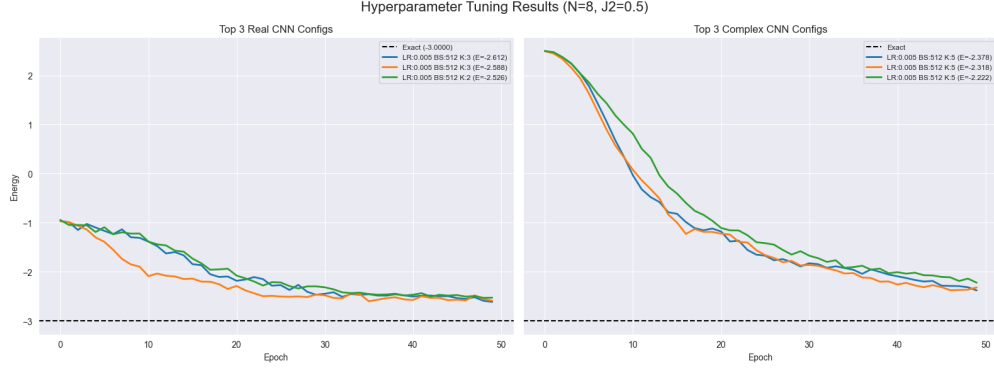


Figure 5: Hyperparameter Grid Search over Depth, Kernel Size

We evaluated the optimization dynamics for a system of $N=8$ spins with frustration $J_2=0.5$ for 500 epochs. As shown in (Fig. 6), the training curves reveals distinct behaviours.

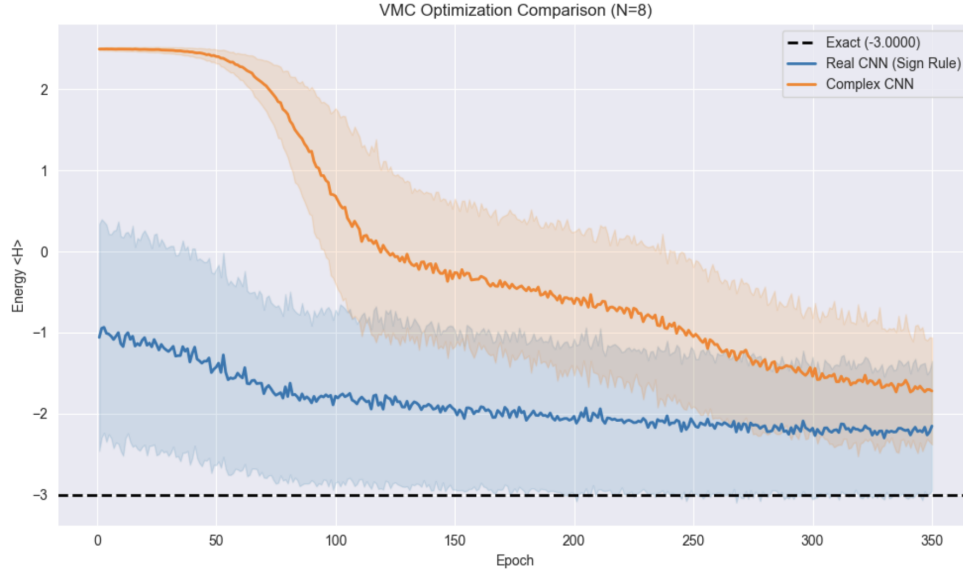


Figure 6: Energy $\langle H \rangle$ vs Epochs

Real valued CNN (Blue line) initializes at a much lower energy (≈ -1.0) and quickly drops to ≈ -2.2 . However, it plateaus significantly above the exact ground state energy (-3.0).

Complex-valued CNN (Orange line) initializes at a high positive energy ($\approx +2.5$). After a "learning plateau" of about 50 epochs, it begins a rapid descent, closing the gap with the Real-Valued model by epoch 350.

Real-valued CNN (Blue line) demonstrates a better initialization due to the hard-coding of the Marshall Sign Rule where the model is able to effectively guess the correct sign structure for the dominant antiferromagnetic interactions. Meanwhile, the Complex CNN struggles with initialization without the physical priors, causing it to perform poorly.

Large- N System

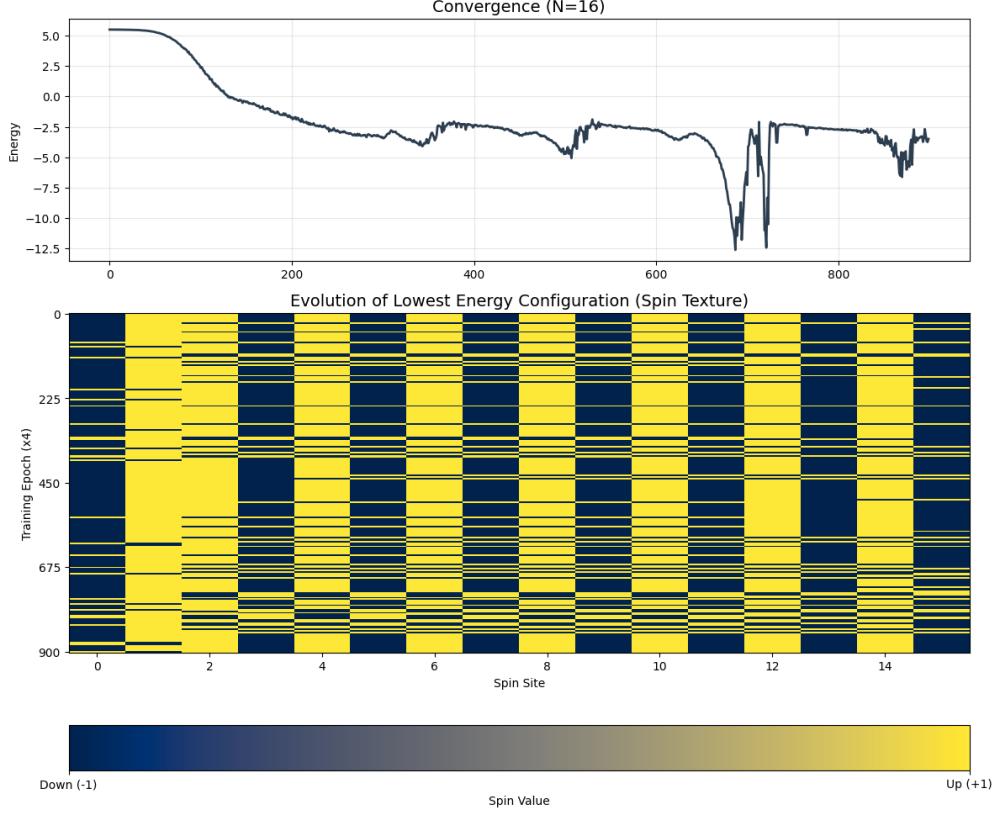


Figure 7: Large N Experiment with optimized hyperparameters

Finally, we extrapolated the optimized Complex CNN to a larger system of $N = 16$ spins, which is impossible to compute exactly. We visualized the "spin texture" (the average magnetization $\langle S_i^z \rangle$ at each site) over the course of training.

As seen in the figure, the spin texture remains largely static over time. This is a positive result as it indicates that the network quickly identified the correct symmetry-broken ground state early in training. Subsequent epochs were spent refining the precise amplitudes of the wavefunction to minimize energy variance.

7 Improvements and Next Steps

While our current models are trained for specific values of J_1 and J_2 , a significant improvement would be the implementation of **Parametric Neural Quantum States**. Instead of fixed constants in the Hamiltonian, we can feed J_1 and J_2 into the network as input features. The network would learn a generalized wavefunction $\Psi(S; J_1, J_2)$. This allows for "Transfer Learning"—the model could sweep the entire phase diagram and identify phase transitions (e.g., from Antiferromagnetic to Dimerized) without needing to be retrained for every new set of coupling constants.

8 Conclusion and Discussion

In this project, we successfully applied Convolutional Neural Networks to approximate the ground states of quantum spin chains. We found that while supervised learning is highly accurate ($\approx 98.5\%$), unsupervised variational learning is significantly more challenging. Our results highlight a trade-off: incorporating physical priors (Marshall Sign Rule) dramatically improves initialization stability but limits the model’s generality in frustrated regimes. Conversely, general complex-valued networks offer greater theoretical flexibility but struggle to navigate the rough energy landscape from a random start.

9 Author Distribution

Zirui (Ann) Zhou: Wrote Introduction, Background of CNN, and Dataset and exploration. Conducted exploratory data analysis.

Qian Yunhan: Coded supervised model and helped with unsupervised model. Wrote supervised model CNN section and parts of unsupervised model CNN section.

Sebastian Seljak: Wrote parts of unsupervised model CNN section, and results section. Helped with complex CNN architecture and energy expectation functions.

Teo Manzo: Wrote the Physical Model and Ground-State Physics section. Helped with preliminary updates to a Stochastic Reconfiguration Algorithm using log-derivatives.

Julieanna Li: Created dataset for CNN. Helped with initial development of supervised and unsupervised CNN.

Zhihao (Justin) Xie: Lead project development and organized project meetings. Developed dataset of CNN. Developed supervised and unsupervised CNN.

10 Github

[Repository](#)

References

- [1] Liang, X., Liu, W.-Y., Lin, P.-Z., Guo, G.-C., Zhang, Y.-S., & He, L. (2018). Solving frustrated quantum many-particle models with convolutional neural networks. *Physical Review B*, 98(10), 104426. <https://doi.org/10.1103/physrevb.98.104426>
- [2] Lange, H., Van de Walle, A., Abedinnia, A., & Bohrdt, A. (2024). From architectures to applications: A review of Neural Quantum States. *Quantum Science and Technology*, 9(4), 040501. <https://doi.org/10.1088/2058-9565/ad7168>

11 Missing/Weaker Components

We believe none of the required components were missing.

11.1 Limitations

The current implementation does not guarantee Hamiltonian consistency (e.g., Heisenberg vs. Ising), nor does it incorporate the J1/J2 couplings with correct physical structures at the model input stage. The sampling updates may not preserve the required quantum subspace, and therefore the variational interpretation is limited. These aspects are left for future verification.

Limitations of this project do not affect the ML demonstration purpose of the course, but they do prevent the current model from being a physically valid NQS/VMC solver.